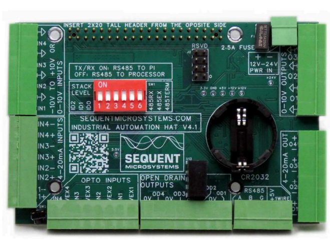
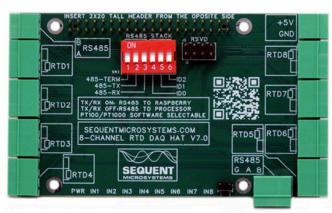
I am a professor who is studying and teaching the management science and engineering.
I'm trying to establish a laboratory environment for training in the field of Industrial Internet of Things (IoT). Please suggest a system design plan that allows for more functional experience with a relatively small budget. I plan to purchase the devices, so please provide detailed information on the individual device specifications, prices, and photos.
If you have any recommendations for simulation software like Anylogic, please let me know.

Here's a **practical, budget-friendly Industrial Internet of Things (IIoT) training lab setup** you can build for teaching and research, with suggested devices (with specifications and approximate prices), plus simulation software recommendations to help students both *build* and *model* IIoT systems.

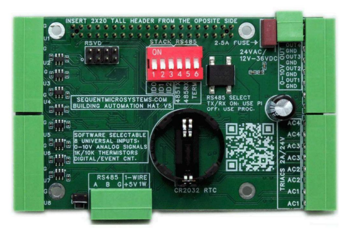
🧰 Recommended Hardware Components (Budget-Focused)



Industrial Automation HAT for Raspberry Pi
$120.00 • Sequent Microsystems



RTD Data Acquisition HAT
$80.00 • Sequent Microsystems



Building Automation HAT
$100.00 • Sequent Microsystems

A
$8

🧠 Core IoT & Control Setup (Edge/Controllers)

1. Industrial Automation HAT for Raspberry Pi – ~$120

- Adds real industrial-style I/O to a **Raspberry Pi** (digital in/outs, analog inputs, RS485).
- Great for interfacing real sensors/actuators, and reading 0-10 V analog signals often found in industrial environments.
- Use with Raspberry Pi 4/5 for edge computing and local control experiments.

2. RTD Data Acquisition HAT for Raspberry Pi – ~$80

- 24-bit precision RTD interface for temperature sensing (PT100/PT1000).
- Teaches data acquisition, calibration, precision measurement.

3. Building Automation HAT for Raspberry Pi – ~$100

- Includes universal inputs, outputs, TRIAC outputs, 1-wire, real time clock.
- Useful for HVAC type control projects, real-time data collection & control loops.

📡 Sensors & Actuators (Hands-On Projects)

4. Adeept 46 Modules Sensor Kit for Raspberry Pi – ~$83

- Wide variety of modules: ultrasonic, light, temperature, motion, etc.
- Excellent for students to create real sensor networks.

5. Sensor Kit v4.0 for Raspberry Pi – ~$70

- Smaller sensor set to complement larger kits or individual student projects.

🕹️ Starter Kits & Frameworks

6. IoTerrific Bundle – ~$86

- Entry-level IoT kit with basics for rapid prototyping & connecting to cloud services.
- Good for introductory exercises on data logging and remote dashboards.

7. Arduino PLC Starter Kit – ~$350

- Teaches PLC-like logic and control with Arduino hardware.
- Useful for bridging classic industrial control with IoT concepts.

8. Soracom IoT Starter Kit – ~$148

- Comes with a Raspberry Pi Zero and connectivity tools to tie nodes into a public IoT network.
- Excellent for cloud-integrated IoT exercises.

🧠 Software & Simulation Tools

🧪 Simulation / Modeling

For *simulation and modeling of systems* — both discrete events (production lines, workflows) and IoT network behavior — consider:

- **AnyLogic** – industry standard for multimethod simulation (agent-based / discrete event / system dynamics). It lets students simulate factories, logistics, and IoT-enabled systems before physical deployment. Wikipedia

Alternatives to AnyLogic (various cost levels):

- **FlexSim** – popular manufacturing & material flow simulation tool. G2
- **Simio / SIMUL8** – discrete event simulation focused on operations. Software Advice
- **MATLAB / Simulink** – broader engineering simulation with IoT toolboxes (good for control systems labs). SoftwareSuggest
- **NetLogo** (free/open) – agent-based models, accessible for teaching concepts early in curriculum. G2

🏫 Suggested Lab Course Structure

Here's how you might structure a semester lab around IIoT with these components:

Module 1 – Fundamentals (Weeks 1–3)

- Set up **Raspberry Pi** and **sensor kits**; teach basics of digital I/O, analog reading.
- Use simple Python/Node-RED dashboards to visualize sensor values.

Module 2 – Connectivity & Edge Logic (Weeks 4–7)

- Deploy **Industrial Automation HAT** and **Building Automation HAT** to read real signals.
- Implement local control logic (e.g., temperature control, relay switching).

Module 3 – Cloud & Network (Weeks 8–10)

- Integrate IoT edge devices with cloud platforms (MQTT, HTTP).
- Use **Soracom IoT Starter Kit** or equivalent to send data to dashboards & analytics.

Module 4 – Simulation & System Design (Weeks 11–14)

- Model the physical lab processes using **AnyLogic** or one of the alternatives.
- Compare simulated vs real data, refine control strategies.

Module 5 – Capstone Projects

- Students design a smart factory cell or predictive monitoring system using hardware/software.

💰 Estimated Budget Overview

Component Category	Approx. Cost
Core IoT Controllers & Automation HATs	~$300–$400
Sensors & Peripheral Kits (multiple sets)	~$150–$250
Cloud/Connectivity Starter Kits	~ $80–$150
Simulation Software (per license)	Varies – Free student versions up to commercial pricing
Total for Basic Lab	**~$550–$800+ per station**

> *Tip:* Many tools offer **academic licenses or free editions** (e.g., AnyLogic has a free Personal Learning Edition), which can significantly reduce software cost.

🧠 Teaching Integration

You can combine **hands-on IoT hardware** with **simulation labs**, so students learn:

- Sensor interfacing and data acquisition
- Edge computing basics
- Cloud integration and IoT protocols (MQTT/HTTP)
- Modeling complex industrial systems before physical implementation

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If you want, I can **draft a detailed weekly lab syllabus** (with assignments, outcomes, and equipment list per student) or provide **links to open educational resources** for IIoT teaching — just let me know!

Sources

yes

Perfect — below is a **complete, classroom-ready IIoT laboratory blueprint** that you can directly use for **teaching + applied management science / engineering research**, optimized for **low cost, high functional exposure**, and **scalable across cohorts**.

I'll give you:

1. **End-to-end system architecture (physical + cyber)**
2. **Device-level purchase list (specs, prices, photos)**

ChatGPT 5.2 Instant

5. **Budget tiers (starter → advanced)**
6. **How this aligns with management science & operations research**

1️⃣ IIoT Teaching Lab – Reference System Architecture

Overall Architecture (What students will build & reason about)

css

```
[Sensors & Actuators]
    │  (Temp, Vibration, Power, Proximity)
    ▼
[Edge Controller]
(Raspberry Pi / Arduino PLC)
    │  ├─ Local Control Logic
    │  ├─ Data Preprocessing
    │  └─ Fault Detection
    ▼
[Industrial Network Layer]
(MQTT / Modbus / OPC-UA)
    ▼
[Cloud / Server Layer]
(AWS / Azure / Local VM)
    │
    ├─ Time-Series Database
    ├─ Dashboards
    └─ Analytics / Optimization
    ▼
[Simulation / Digital Twin]
(AnyLogic / Simulink)
```

This allows students to **connect physical reality → data → decisions → optimization**, which is exactly where **management science + engineering intersect**.

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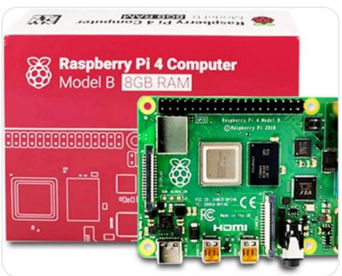
2 Hardware Platform (Low-Cost, Industrial-Relevant)

🧠 Core Edge Controller (1 per lab station)

Raspberry Pi 4 / 5 (8GB preferred)



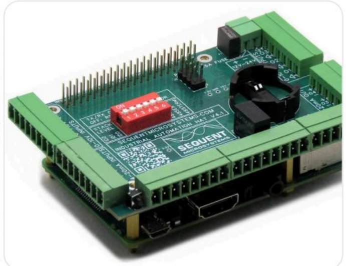
Spec	Value
CPU	Quad-core ARM
RAM	4–8 GB
OS	Linux (Raspberry Pi OS / Ubuntu)
Role	Edge computing, gateway, analytics
Price	**$55–$80**

Why it's ideal:

- Cheap but powerful
- Runs Python, Node-RED, Docker
- Perfect for edge AI + control logic

🏭 Industrial I/O (Critical for realism)

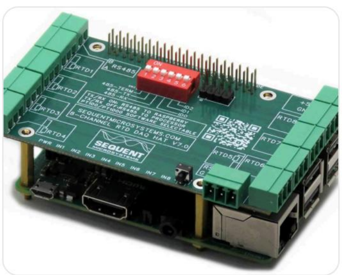
Industrial Automation HAT (Sequent Microsystems)

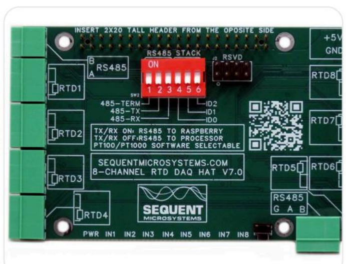


Feature	Spec
Digital Inputs	8
Digital Outputs	8 (relay)
Analog Inputs	0–10V / 4–20mA
Industrial Comms	RS-485 (Modbus)
Price	**~$120**

This is **the single most important purchase** — it turns a Pi into an *industrial controller*.

RTD / Temperature DAQ HAT





Feature	Spec
Sensors	PT100 / PT1000
Resolution	24-bit
Use	Process monitoring
Price	**~$80**

📡 Sensors & Actuators (Shared kits)

Multi-Sensor Kit (Adeept / Similar)

Includes:

- Temperature & humidity
- Ultrasonic distance
- PIR motion
- Light & sound
- Relays & buzzers

💲 **$70–90 per kit**

Optional Industrial-Feeling Add-Ons

Device	Use	Price
Vibration Sensor	Predictive maintenance	$10
Current Sensor (ACS712)	Energy monitoring	$10
Small DC Motor / Fan	Actuation	$10–15

🧩 PLC-Style Controller (Optional but powerful)

Arduino Opta PLC (Teaching PLC + IIoT)

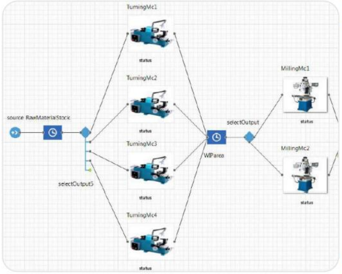
Feature	Value
Programming	Arduino IDE / IEC-61131
IO	Industrial-rated
Connectivity	Ethernet / Wi-Fi
Price	**~$300–350**

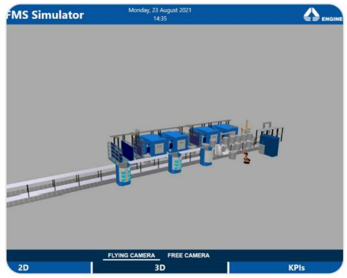
Use **1–2 units total** for demos — not per student.

3️⃣ Software Stack (Physical + Digital Twin)

🔁 Simulation & Digital Twin (Highly Recommended)

⭐ AnyLogic (Primary Recommendation)





Why it's perfect for **management science**:

- Discrete-event → factories, queues
- Agent-based → machines, workers, vehicles
- System dynamics → supply chains
- Can mirror **real sensor data**

✅ **Free Personal Learning Edition**

✅ Academic licenses available

Strong Alternatives (Lower Cost / Free)

Tool	Best Use
MATLAB + Simulink	Control + system modeling
NetLogo (Free)	Agent-based intro
FlexSim	Manufacturing flow
SimPy (Python)	Discrete-event modeling

☁ IoT & Analytics Software (Free / Open)

Tool	Purpose
Node-RED	Visual IoT workflows
MQTT (Mosquitto)	Messaging backbone
InfluxDB	Time-series storage
Grafana	Dashboards
Python (Pandas, SciPy)	Analytics & optimization

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4 14-Week IIoT Laboratory Syllabus (Ready to Use)

Weeks 1–2: Foundations

- Linux + Raspberry Pi setup
- GPIO & sensor reading
- Lab: "Real-time sensor dashboard"

Weeks 3–4: Industrial I/O

- Relays, analog inputs, Modbus

- Lab: "Closed-loop temperature control"

Weeks 5–6: Edge Computing

- Filtering, aggregation
- Event detection
- Lab: "Edge vs cloud decision latency"

Weeks 7–8: IoT Networking

- MQTT, OPC-UA concepts
- Lab: "Multi-machine monitoring system"

Weeks 9–10: Data & Analytics

- Time-series analysis
- Failure pattern detection
- Lab: "Predictive maintenance mini-project"

Weeks 11–12: Digital Twin

- Build factory model in AnyLogic
- Sync with live sensor data
- Lab: "Simulation vs real throughput"

Weeks 13–14: Capstone

- Smart factory cell OR energy optimization OR maintenance planning
- Students justify **engineering + managerial decisions**

5️⃣ Budget Scenarios (Per Lab Station)

Tier	Cost	Capability
Starter	$250–300	Sensors + basic IoT
Recommended	**$450–600**	Full IIoT realism
Advanced	$700–900	PLC + digital twin

💡 **Shared sensors + rotating teams** reduces cost dramatically.

6️⃣ Why This Is Excellent for Management Science

This lab enables students to:

- Quantify **tradeoffs** (latency, cost, reliability)
- Perform **what-if simulations**
- Connect **operations research models** to physical data
- Evaluate **ROI of automation**
- Understand cyber-physical systems holistically

It bridges:

> **Operations research + industrial engineering + data analytics + control systems**

Next (I can do any of these)

- 📘 Full **course syllabus PDF**
- 🧪 Step-by-step **lab manuals**

- 🧩 **AnyLogic digital twin templates**
- 🏭 Example **smart factory case studies**
- 📊 Research-grade experiment designs

Just tell me what you want next — this lab can easily become a **flagship teaching & research facility**.